

COURSE 3 - DATA ENGINEERING 2026**C3-102 - RetailPulse Protected Review**

Diagnostic review. Open the matching Rxx only after the Pxx attempt exists.

Artifact	C3-102
Status	2026 - LEARNER EDITION
Phase	C3-09 Production Capstone & Engineering Transition 15 lessons 36-48 focused hours
Prerequisite	C3-08 Gate PASS; all earlier C3 evidence retained.
Assessment boundary	C3-104 is the phase Gate only. C3-113 remains the later course-wide Final Production Gate.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: requirement -> mental picture -> small proof -> guided integrated build -> expected result -> why it worked -> break/fix -> changed independent case -> validation -> production reasoning -> evidence. Integration is not mastery until the learner can explain and recover the system independently.

Review rule

This pack is not a solution manual to copy. Diagnose the first weak contract/control, close the review, then complete a changed retry without reproducing the reviewed wording/code.

A review never deletes the first attempt. Passing evidence must survive after a changed retry.

R01 - Review Capstone ADR: business problem, architecture and non-functional requirements

Expected engineering shape

ADR names users/data products, measurable NFRs, at least two options, chosen design, rejected alternative and consequences. Tool names are secondary to requirements.

Look for this first

If the ADR says only which tools you like, restate the consumer outcome and failure budget before choosing architecture.

Common failure patterns

- Starting with Kafka/Spark/Fabric names before writing the business outcome.
- Writing vague NFRs such as "fast" or "secure".
- Listing one option only, so no decision was actually made.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Change the requirement from daily to sub-minute fraud detection and state exactly which architectural components must change.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R02 - Review Source contracts, synthetic/public data and reproducible inputs

Expected engineering shape

Contracts state explicit grains/keys/change semantics/ownership and distinguish compatible versus breaking evolution. No employer/private data is needed.

Look for this first

If you cannot define a stable key or change timestamp, the incremental design is not ready. Record the limitation rather than inventing CDC.

Common failure patterns

- Using filename as business identity.
- Calling max timestamp "CDC" without source guarantees.
- Using private employer data in a public portfolio.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Add a source field that is optional today but required by a future consumer; define migration and compatibility behavior.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R03 - Review Build the batch ingestion path

Expected engineering shape

Local run reports status success, 7 current rows, completed revenue 6910.0 and reconciled=true; source-preserving landing metadata contains hashes/run identity.

Look for this first

If rerun doubles output, separate immutable landing evidence from key-based current-state publication. If source bytes change under the same name, fail or version them explicitly.

Common failure patterns

- Overwriting source evidence.
- Using append-only publication for current-state data.
- Treating duplicate runs as duplicate business records.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Rerun the same base file twice, then rename it without changing bytes; explain identity behavior.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R04 - Review Add incremental / CDC-style change handling

Expected engineering shape

O102 is 1050.00, O106 and O107 each appear once, current row count is 7, completed revenue remains 6910.0 after replay.

Look for this first

If an incremental replay changes totals again, your update is not idempotent. If two records tie on change time, declare a deterministic tie-break rule.

Common failure patterns

- Advancing watermark before validated publish.
- Ignoring late/out-of-order corrections.
- Using nondeterministic tie handling.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Apply a late correction whose updated_at predates the last observed watermark and explain how your source contract prevents or handles loss.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R05 - Review Design layered lakehouse and serving architecture

Expected engineering shape

Every layer has a different responsibility/trust contract; serving choice is tied to a real consumer pattern; no layer exists only for naming convention.

Look for this first

If every layer contains the same data, explain what trust/transformation boundary is missing. If Warehouse exists only because it sounds enterprise, remove it or justify the SQL consumer.

Common failure patterns

- Creating Bronze/Silver/Gold copies with no distinct contract.
- Choosing Warehouse only because it sounds enterprise.
- Mixing raw replay retention with curated serving access.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Redesign for a second consumer that needs raw event history instead of current-state orders.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R06 - Review Transformation engineering, modeling and tests

Expected engineering shape

Tests run from a clean project with ``python -m pytest -q``; current-state grain and daily sales are independently validated; changed-rule impact is bounded and explained.

Look for this first

If a test merely asserts the same logic as the transform, add an independent invariant or manual control. If totals change after a refactor, fail closed.

Common failure patterns

- Testing only that code executes.
- Writing a test that copies the same transformation logic.
- Hiding duplicate/grain problems with DISTINCT.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Change one business rule for completed revenue and prove the expected result plus regression boundary.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R07 - Review Orchestration, retries, backfills and partial reruns

Expected engineering shape

DAG/design distinguishes retryable vs deterministic failure, scopes work by data interval, supports one bounded backfill and proves partial rerun does not duplicate/corrupt state.

Look for this first

If every failure retries, classify deterministic data defects separately. If backfill overwrites all history, restore partition/date-scoped publication.

Common failure patterns

- Retrying deterministic data defects.
- Backfilling all dates to repair one partition.
- Clearing tasks without understanding side effects.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Backfill one historical service date after changing only its source file and prove unrelated dates are unchanged.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R08 - Review Integrate a real Spark component

Expected engineering shape

Spark output matches the local reference on agreed keys/counts/values; scale note explains why Spark is not the cheapest choice for the tiny fixture but what changes at larger scale.

Look for this first

If Spark output disagrees with the reference, fix correctness before tuning. If the dataset is tiny, state that Spark is scale-transfer evidence rather than a cost-optimal local choice.

Common failure patterns

- Using Spark on tiny data and calling it "big data".
- Tuning before parity is correct.
- Claiming scale numbers without measured evidence.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Explain the first bottlenecks/partition changes at 10x and 100x volume without inventing a cluster benchmark.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R09 - Review Quality gates and source-to-output reconciliation

Expected engineering shape

Quality gate blocks injected corruption even when row count alone could look correct; post-repair keys/values/totals reconcile exactly.

Look for this first

If row counts match but values differ, compare key sets and record-level values. If totals match but keys differ, treat total parity as insufficient.

Common failure patterns

- Checking only row count.
- Using the same derived result as its own reconciliation.
- Allowing warning-only checks for critical integrity invariants.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Create an offsetting defect that preserves the grand total and prove the key/value reconciliation still catches it.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R10 - Review Observability, SLOs and operational monitoring

Expected engineering shape

SLIs/SLOs are measurable, run evidence is correlated, alerts have owner/action/runbook, and page-worthy versus ticket-worthy conditions are justified.

Look for this first

If an alert has no owner/runbook or fires on every harmless anomaly, redesign it around consumer impact and actionable evidence.

Common failure patterns

- Alerting on every anomaly.
- No run_id/correlation across logs.
- SLO defined from an internal metric with no consumer impact.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Change the business cutoff/SLO window and explain the alert effect without changing pipeline code.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R11 - Review Security, governance and privacy-safe evidence

Expected engineering shape

Static secret scan is clean; permissions are least-privilege by role; one deny boundary is specified/proven where runtime permits; portfolio evidence contains no private secrets/PII.

Look for this first

If a workflow needs broad write/admin access for convenience, reduce scope. If evidence contains real secrets/PII, rotate/revoke if needed and rebuild sanitized evidence.

Common failure patterns

- Putting secrets in screenshots/logs.
- Testing data restrictions as an admin.
- Granting broad roles just to make a demo work.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Add a contractor who can read curated output but must not read raw/landing data; show the control-plane/data-plane distinction.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R12 - Review CI/CD, deployment and controlled change

Expected engineering shape

CI fails closed on tests/checks; deploy is protected; credentials are not unnecessarily long-lived; post-deploy smoke includes data reconciliation, not only workflow success.

Look for this first

If deploy can run when tests fail, restore dependency. If prod credentials are available before approval, fix the environment boundary.

Common failure patterns

- Deploy job can run after failed tests.
- Production secrets accessible before approval.
- Calling successful deployment proof of data correctness.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Introduce a schema/business-rule change and specify rollback versus roll-forward plus post-change reconciliation.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R13 - Review Incident injection, recovery and prevention

Expected engineering shape

Incident evidence shows defect detection -> containment -> root cause -> bounded repair/replay -> post-recovery reconciliation -> prevention update.

Look for this first

If recovery deletes/rebuilds everything without need, reduce blast radius. If you cannot prove the repaired result, the incident remains open.

Common failure patterns

- Damaging canonical evidence to create a demo.
- Deleting/rebuilding everything instead of bounded recovery.
- Closing incident before post-recovery reconciliation.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Repeat with a different defect class such as missing customer FK or duplicate order key.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R14 - Review Runbook, README, handoff and portfolio evidence

Expected engineering shape

A fresh reviewer can set up, run, test, troubleshoot and understand evidence using only README/runbook; claims are linked to sanitized proof and limitations are explicit.

Look for this first

If another engineer needs your memory to run the project, document hidden prerequisites. If a claim is not backed by evidence, weaken or remove it.

Common failure patterns

- README is marketing only.
- Runbook lacks exact success/failure signals.
- Portfolio claims live tools that were never executed.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Hand the repository to a fresh reviewer using only README/runbook and record every missing instruction.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.

R15 - Review Final architecture defense and engineering transition

Expected engineering shape

Defense explains every major component, failure/recovery/evidence path, vendor-specific boundary and 10x/100x change. C3-104 is understood as phase Gate, not graduation.

Look for this first

If you answer with tool definitions instead of system behavior, return to requirements, data flow, failure mode and evidence. Do not claim live cloud/Spark/Airflow execution you did not perform.

Common failure patterns

- Answering with definitions instead of data flow/failure reasoning.
- Claiming benchmark/production experience not evidenced.
- Confusing C3-104 phase pass with C3-113 graduation.

Changed retry

Do not repeat the original prompt unchanged. Use this transfer: Defend a variant where Fabric is replaced by another lakehouse/cloud platform and separate permanent engineering principles from vendor features.

Pass evidence

Changed artifact/output + independent validation + one recovery note + 60-90 second explanation. If live heavy-tool evidence is unavailable, record the precise blocked runtime and do not claim execution.